JOSEPH GRUBER

Space & Satellite Systems Engineering Manager

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U.S. Citizen

Secret (Inactive)

SUMMARY

Passionate, mission-driven systems engineer and engineering manager with over 15 years of system engineering, program management, and engineering leadership experience. From designing, testing, and deploying a complex autonomous system for satellite constellation mission management to turning around a team that provided high-impact validation for a spaceflight vehicle, my background spans across all mission lifecycle phases in the defense and space industry.

I bring broad cross-functional expertise in complex programs, spacecraft mission operations, integration & testing (I&T), verification & validation (V&V), project management, and anomaly resolution. I excel at leading multi-disciplinary teams in combining modern systems & software engineering best practices with an 'agile aerospace' mindset and a strong bias for action to solve today's space industry's engineering challenges.

EXPERIENCE

Amazon Web Services – Denver, CO — Jun 2020 – Present

AWS Ground Station – Sr. Technical Program Manager

- Drove agile software development efforts across cross-functional teams of software engineers, network engineers, and RF/antenna engineers. Developed analytics to automate and improve operational efficiency.
- Managed schedules, specified deliverables, and defined roadmaps to scale and sustain operations and reliability of cloud-based ground station network for satellite commanding. Supported anomaly investigation and recovery for high-severity events.
- Created launch and early orbit phase (LEOP) operational procedures. Led test events & simulations to ensure mission readiness. Orchestrated LEOP's support team for the deployment of numerous satellites across multiple satellite operators.
- Executed efforts to design, build, validate, and deploy capability for custom ephemeris data (TLE, OEM) support for scheduling satellite contacts.
- Planned and coordinated the software launch of three new international, multi-antenna ground stations and two new AWS ground station regions.
- Provided technical subject matter expertise of spaceflight operations to product managers, solution architects, and others within the tech industry.

Tech Stack: AWS (Lambda, EC2, etc.), Python, Orekit, STK

Blue Origin – Kent, WA — Feb 2019 – Jun 2020

New Shepard Production Test & Operations – Technical Project Manager

- Managed a dedicated team of 10+ software engineers developing test software to validate and qualify operational human-rated flight hardware.
- Developed and reviewed multiple test scripts for qualification and acceptance testing in coordination with the test engineering team.
- Drove improvements to software development processes in alignment with mission and safety-critical engineering standards.

EDUCATION

Embry-Riddle Aeronautical Univ.
Masters of Aeronautical Science (Space Studies)

Florida Institute of Technology
B.A. Business Administration (Computer Information Systems)

Hillsborough Community College
A.A. Computer Information Systems

CERTIFICATIONS

AWS Solutions Architect - Associate

FreeFlyer - Level 1

Project Management Professional (PMP)

Professional Scrum Master (PSM-I)

Satellite Tool Kit (STK) - Master

AWARDS

NASA Flight Dynamics Support Services Team Technical Innovation Award

NASA/NOAA GOES-R Process Improvement and Innovation Award

New Shepard Software Architecture – Technical Project Manager

- Guided a team of 25+ software, network, GNC, and system integration engineers through design, development, and verification & validation of sub-orbital human-rated launch vehicle flight software.
- Led process improvement effort and coached team leads to reduce flight software development time resulting in the first on-schedule flight software mission deliverable.
- Organized creation of multiple COTS software compliance plans documenting safety and reliability for software qualification review.
- Created automated collection and reporting of scope metrics and requirements verification via DOORS, improving communication and transparency to executive leadership and founder.
- Drove implementation with operations & maintenance team for continuous testing and deployment of flight software and integrated ground system.
- Documented and instituted workflows to manage scope for multiple payload and test missions coordinating parallel software release schedules.

Tech Stack: C++, Python, MATLAB, Jenkins, Gitlab, AWS, Ubuntu

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a.i. solutions – Lanham, MD — Jul 2016 – Feb 2019

NASA Space Network Ground Segment Sustainment (SGSS) – Consulting Systems Engineer

- Analyzed system architecture design as technical lead for upgraded TDRSS ground terminals and advised on engineering best practices for modernization and future sustainment operations of the space network operations center.
- Formulated lifecycle supportability risks and associated mitigation strategies from contractor system design and architecture documentation in preparation for operational handover post-final acceptance review.
- Created a configuration control system for long-term asset planning reporting for several thousand CSCI's.

Tech Stack: Oracle Database, WebLogic, & Application Server, VMware

NASA Earth Science Mission Operations (ESMO) – Mission Systems Engineer & Functional Manager

- Managed, mentored, and developed a team of 20+ multi-disciplinary system and software engineers supporting flight dynamics of multiple NASA satellites and satellite situational awareness (SSA) of international earth science satellite constellation.
- Designed and deployed CI/CD pipelines for flight dynamics software and constellation coordination software resulting in an 80% reduction of manual testing.
- Planned and implemented automated close approach violation and debris avoidance maneuver capabilities through the automated acquisition of ephemeris data allowing for a 5x increase in daily conjunction assessments.
- Spearheaded process improvements and implemented corrective actions achieving CMMI-DEV Level 3 maturity.
- Created migration plan and led initiative to upgrade and modernize legacy, end-of-life software components ensuring reduced risk during the increased lifespan of multiple spacecraft.
- Developed and executed a $2.5 million project management plan with a consistent annual ~10% revenue growth.

Tech Stack: Node.js, C++, C#, Python, TeamCity, Bitbucket, FreeFlyer, VMware, Hyper-V, MATLAB

Hawaii Space Exploration Analog & Simulation (HI-SEAS) — April 2013 – June 2018

Mission Support Lead (Research/Unpaid)

- Planned and coordinated mission operations for multiple analog missions in a simulated Martian habitat.
- Recruited and guided 30+ mission operations team members on long-duration, human spaceflight simulations as the primary communications and support interface to isolated crew members.
- Developed frontend and backend services on AWS using Python and PHP for proto-flight distributed monitoring of habitat systems.
- Architected unique time-delayed communication protocols to simulate deep space communication delays.

ECG, Inc. – Greenbelt, MD Mar 2014 – Jan 2016

NASA Geostationary Operational Environmental Satellite R-Series (GOES-R) – Lead Planner & Team Lead

- Responsible for schedule and technical coordination during the design, assembly, integration & test of spacecraft, instrument, and ground segment schedules and statistical risk analysis of program scope.
- Contributed to architecture and development of petabyte-scale data storage solution for CCSDS-formatted level zero raw products used for spacecraft instrument calibration and validation.
- Generated plans and timelines for launch and orbit raising activities to validate on-orbit performance requirements and mission data products' operational readiness.
- Supported mission, flight, and launch readiness reviews on behalf of the chief engineer's office, creating and organizing schedule analysis and performance analysis reports.

Wyle Laboratories – Arlington, VA Feb 2012 – Mar 2014

F-35 Joint Strike Fighter (JSF) – Lead Planner & Team Lead

- Led planning team for initial operational test & evaluation phase providing programmatic and technical oversight throughout integrated baseline reviews, system design, and development flight testing.
- Conducted probabilistic risk analysis using Monte Carlo simulations to quantify technical and programmatic risks.
- Developed custom Python solutions for automated data analysis and metrics reporting.
- Witnessed and supported test events, simulations, and validation exercises, reporting results and metrics.

Early Career

- Systems Integration Engineer – United States Southern Command (USSOUTHCOM), SAIC & CTSC
- Systems Engineer – United States Central Command (USCENTCOM), Lockheed Martin

PROFESSIONAL AFFILIATIONS

- Consultative Committee for Space Data Systems (CCSDS): Associate
- PMI; PMBOK 5th Edition: Risk Management Content Editor
- American Institute of Aeronautics and Astronautics (AIAA): SciTech Conference Chair
- AIAA National Capital Section (NCS): Communications Committee
- International Council on Systems Engineering (INCOSE)
- Toastmasters: President, Advanced Communicator, Advanced Leader

RELEVANT SKILLS

Platforms

Amazon Web Services (Compute, Database, Networking, Storage), Windows Server, Linux (Ubuntu, Red Hat), RTLinux, Mac OS X, Solaris, VMware, NASA core Flight System (cFS)

Software

a.i. solutions FreeFlyer, AGI Satellite Toolkit (STK), Atlassian (JIRA, Confluence, Bitbucket), GitHub, Gitlab, IBM Rational DOORS, IBM Rational Team Concert (RTC), Jenkins, Microsoft Project, Microsoft SQL Server, MySQL, MongoDB, MathWorks MATLAB, MathWorks Simulink, Oracle Primavera P6, Oracle Primavera Risk Analysis (PRA), OS/COMET, TeamCity

Programming Languages

Python, JavaScript, Node.JS, PHP, C#, C++, SQL, Visual Basic.Net, Visual Basic for Applications (VBA), Bash

Mechanisms & Process

Agile methodology (Scrum, Kanban), Schedule Management, Test Driven Development, Automated Testing, Requirements Development, Release Planning, NPR 7150.2, NPR 7120.5, DO-178C, CMMI-DEV (Level 3)